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Shining light on memory

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Research report

Shining light on memory: Effects of bright light on working memory performance

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HIGHLIGHTS

- We discuss a neurobehavioural view on non-image forming performance effects of light.
- Based on this view, we examined whether these effects depend on task difficulty.
- Illuminance level differentially affected easy vs. difficult digit-span performance.
- In the afternoon, bright light led to worse performance on two relatively difficult tasks.
- Bright light improved performance on the easier digit-span task in the afternoon.

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ABSTRACT

This study examined whether diurnal non-image forming (NIF) effects of illuminance level on cognitive task performance depend on task difficulty and time of day. We employed a balanced crossover design with two 60-min sessions of 200 vs. 1000 lux at eye level. Digit-span task difficulty was manipulated within subjects (forward (FDST) vs. backward (BDST) digit-span task), *n*-back task difficulty was manipulated between subjects ($n = 1$, 2, or 3). Bright light exposure improved FDST performance during the final measurement block, especially in the afternoon. In contrast, BDST performance deteriorated slightly under bright light in the afternoon. Two-back performance was significantly worse under bright light in the afternoon, while no effect of illuminance level was found on 3-back performance. Thus, the more difficult BDST was affected differently by light intensity as compared to the easier FDST. *N*-back accuracy, however, did not confirm this role of task difficulty. Future studies should investigate whether similar results hold for other types of tasks and how other variables (e.g., time of day, physiological arousal, or other task characteristics) may influence the direction and magnitude of NIF effects on performance.
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1. Introduction

Via the non-image-forming (NIF) pathway, light can affect the circadian rhythm and induce acute alerting and activating effects throughout the 24-h day [1,2]. It is now well established that a third class of retinal photoreceptors, the so-called intrinsically photosensitive retinal ganglion cells (ipRGCs), are primarily responsible for these NIF effects of light [3–5]. A substantial amount of research has shown that high (as compared to low) illuminance levels can exert acute physiological as well as subjective alerting effects during the biological night [6–11] and day [10–14]. These acute alerting effects of bright light sometimes, but not always, translate into enhanced sustained attention and cognitive performance abilities [6,10–13,15,16]. On occasion even performance undermining effects are reported (e.g., [16]). The mechanisms that may explain these inconsistent effects of illuminance level on task performance are still unknown. Yet, if we aim to optimize cognitive performance through light intensity, it is essential to better understand these processes. Therefore, the focus of the current study was to further investigate the effects of illuminance level on cognitive task performance.

1.1. Cognitive performance under bright vs. dim light

During the biological night, exposure to bright light – as compared to dim light – either improves sustained attention and cognitive abilities, or leaves them unaffected [6,11,15,17–20]. For

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example, neither Campbell et al. nor Kretschmer et al. found an effect of bright vs. dim light on sustained attention, but they did report improvements on working memory and arithmetic abilities [15] as well as on logical reasoning [18]. In contrast, Boyce et al. did not find improvements on logical reasoning abilities but did report improvements on digit recall and grammatical reasoning [17]. Overall, no consistent pattern of bright light effects on task performance can be found in these studies. However, it should be noted that the lighting characteristics (such as illuminance level and correlated colour temperature (CCT)) as well as the timing and duration of the light exposure are quite different from one study to the other. This may partly explain why light exposure sometimes does, and at other times does not show positive effects on cognitive performance.

Fewer studies have focused on NIF effects of light on cognitive performance during the biological day. Those that have, however, revealed even more contrasting results, including not only positive and null effects, but also negative ones [11–14,16,21,22] Although Smolders et al. [13,16] and Phipps-Nelson et al. [12] reported improvements on sustained attention under bright light, results on a task requiring visual scanning revealed mixed effects and results on working memory and inhibitory abilities even showed performance-undermining effects of bright vs. dim light exposure [13,16]. Rüger et al. [11], on the other hand, showed performance improvements under bright light on visual search and arithmetic abilities, but no improvements on sustained attention. Again it should be noted that light exposure characteristics differed substantially between studies. In addition, subjects were light and/or sleep deprived prior to the light manipulation in the studies of Phipps-Nelson et al. [12] and Rüger et al. [11], which may have influenced subsequent effects of light exposure on cognitive task performance.

All in all, it can be concluded that the collective results of studies examining the NIF effects of light intensity on cognitive task performance are quite equivocal. Possible explanations for these contrasting findings may be partly found in differences in lighting characteristics (e.g., illuminance level or CCT) or timing characteristics (timing and duration of the light exposure) (see [1] for a review). However, there are also studies reporting differential effects of illuminance level on task performance within one study paradigm [6,11,16,17,21]. In these studies, various tasks were performed under the same light manipulation, but differences in the direction and magnitude of NIF effects of light on cognitive performance were found. These results suggest that task characteristics may play an important role in the effect illuminance level on cognitive performance.

1.2. Mechanisms underlying daytime performance effects

Recent neuroimaging studies provide evidence for mechanisms that may explain diurnal acute alerting effects of light, as well as light's differential effects on cognitive task performance [14,23,24]. These studies indicate that bright and/or monochromatic blue light exposure modulates activation of brain areas related to alertness and regulation of physiological arousal levels. More specifically, these results revealed light-induced activation in subcortical brain areas related to bodily arousal regulation such as the thalamus and the locus coeruleus (LC) [14,25]. Activation of the LC may in turn affect bodily state arousal levels via changes in the noradrenergic system [26]. These changes in arousal level have been found to influence task performance as described by The Yerkes–Dodson Law (YDL) [27]. The YDL explains that the relationship between arousal levels and task performance may be moderated by task difficulty. It suggests that for easier tasks (i.e. tasks needing only focused attention on a restricted range of stimuli), performance improves with increasing arousal levels in a dose-dependent manner following a logistic function. However, for more difficult tasks (i.e. tasks requiring divided attention (multitasking) and/or higher executive functions) an inverted U-shape relationship between performance and arousal was found. This relationship showed lower performance levels under low (i.e. when a person feels sleepy, drowsy and/or bored) and high (i.e. when a person is stressed or anxious) arousal levels, and optimal performance levels under intermediate arousal levels.

In this line of reasoning, previous research revealed that, as compared to relatively low illuminance levels ($\leq$500 lux at eye level), high illuminance levels ($\geq$1000 lux at eye level) increase physiological arousal levels as measured by heart rate, skin conductance and muscle sympathetic nerve activity [10,28,29]. Therefore, it is possible that acute light-induced activation of the brainstem (especially the LC) affects cognitive performance via changes physiological arousal levels. However, although the YDL is commonly accepted and empirical support for the YDL does exist [30–33], it is also criticized on methodological issues in the original study as well in later studies replicating the YDL. Points of discussion are the precise conceptualization of arousal and how to properly change physiological arousal levels in human subjects [34–37]. Therefore, it can only be speculated that illuminance level affects cognitive performance in a task-dependent manner by changing arousal levels.

The studies of Vandewalle et al. [24] also revealed light-induced brain activation in cortical areas related to attention regulation such as the dorsolateral prefrontal cortex, the intraparietal sulcus (IPS) and the superior parietal lobe [14,23,38]. Moreover, they found light-induced activation in cortical areas related to higher executive functioning and working memory abilities such as the IPS, the middle frontal gyrus and the supramarginal gyrus [25,38]. These increases in brain activation under bright and blue light were found during performance on a working memory task. However, although changes in brain activation in regions related to arousal and cognitive functioning were observed, these studies did not reveal differences in participants' actual cognitive performance abilities under the different lighting conditions. It therefore remains unclear whether light-induced brain modulation may (partly) explain differences in performance under different lighting conditions. A limitation of these neuroimaging studies is that the light exposure duration was relatively short (maximum of 20 min). Vandewalle et al. [24] hypothesized that modulation of brain networks may temporarily enhance cognitive performance when the duration of the light exposure period is extended. Obviously, additional studies employing longer light exposure durations are needed to investigate whether changes in brain activation can explain light-induced changes in cognitive performance.

1.3. The current study

It can be concluded that there is considerable evidence that illuminance level can exert acute NIF effects on alertness and performance during the biological day. However, findings regarding NIF effects of light on cognitive task performance are quite inconclusive and the underlying mechanisms through which these effects are manifested are largely unknown. As explained previously, the effect of physiological arousal on task performance may be moderated by task difficulty. Because bright light exposure has been found to enhance physiological arousal levels [10,28,29], optimal light exposure for cognitive task performance may depend on task difficulty. That is, optimal performance levels for more challenging tasks are expected to occur at lower illuminance levels (lower physiological arousal) as compared to optimal performance levels for easier tasks. In order to investigate this hypothesis, we compared cognitive performance under two illuminance levels (200 and 1000 lux at eye level) on two working memory tasks, which were manipulated in difficulty level. In addition to task

performance, subjective performance, sleepiness, vitality, tension and mood were assessed. Furthermore, we examined time of the day as a potential moderator in the analyses as it is known that cognitive performance abilities vary over the day based on a person's circadian phase and homeostatic sleep pressure [39,40]. In addition, previous findings revealed that bright light exposure during the biological day may lead to more beneficial effects in the morning than in the afternoon [13,41]. Based on the current state of knowledge regarding possible underlying mechanisms of light-induced NIF effects, we hypothesized that a higher illuminance level would be more beneficial for performance on relatively easy tasks as compared to more difficult tasks. Moreover, performance on difficult tasks may be unaffected or even decreased by a higher illuminance level. In addition, we expected beneficial effects of a higher illuminance level on self-reported vitality, sleepiness and mood. These effects were hypothesized to be more pronounced during the morning than the afternoon hours.

2. Methods

2.1. Design

This study employed a mixed balanced crossover design to investigate the effect of illuminance level (200 lux vs. 1000 lux at eye level) on cognitive performance as a function of task difficulty. Two types of tasks were employed, namely the digit-span task and the *n*-back task. Digit-span difficulty was manipulated within subjects. *N*-back task difficulty and time of day were manipulated between subjects. Performance on the *n*-back and two digit-span tasks as well as subjective indicators were measured during the baseline phase (100 lux et eye level) and in four repeated measurement blocks during the experimental phase in which illuminance level was either 200 or 1000 lux at eye level.

Participants came to the lab on two separate days (with at least two days in between sessions) at the same time of the day (9:00–10:30 AM; 10:45 AM–12:15 PM; 12:15–13:45 PM; 13:45 PM–15:15 PM or 15:45–17:15 PM). Participants were randomly assigned to one of the three *n*-back difficulty conditions, the order of the lighting manipulation, and their work station. There were two versions of each task (one for each session). Task version order was counterbalanced across participants.

2.2. Participants

Sixty-four participants (mean age = 21.4; *SD* = 2.1; 32 male, 32 female) completed both light exposure sessions. About half of the participants was first exposed to the 1000 lux condition (*N* = 33) while the other half was first exposed to the 200 lux condition (*N* = 31).

Participants were recruited at the Eindhoven University of Technology in the Netherlands via advertisements, social networks and the university's participant database. Before participation, participants were screened using the Munich Chronotype Questionnaire [42]. Based on the Midpoint of Sleep on Free days corrected for accumulated sleep pressure during working days (MSFsc) found for subjects aged 18–30 years in the Netherlands (see [43]), extreme chronotypes (falling outside the 25–75% range for this age category) were excluded from participation. After approval, participants were instructed to register on two similar timeslots on regular weekdays with at least two days in between sessions. The two sessions were separated by eight days (*SD* = 6.0) on average. The distribution of participants across n-back versions and timeslots is presented in Table 1.

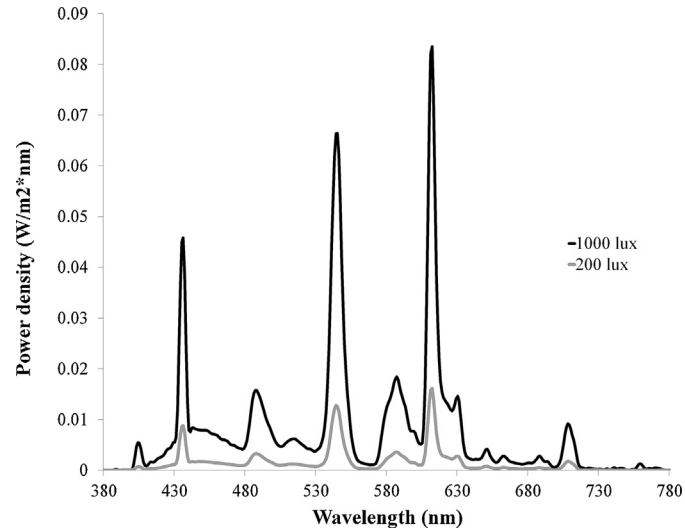

Fig. 1. Spectral power distribution measured at eye level in the 200 lux and 1000 lux (4000 K) condition.

2.3. Setting

The laboratory room setting was a simulated office environment at the Eindhoven University of Technology with a size of 3.9 m by 7.4 m. Four workstations were created in this room, separated by a curtain and off-white office panels. Participants could not see each other during the sessions.

The laboratory room was equipped with recessed Philips Savio luminaires in the ceiling. Each ceiling luminaire (Philips Savio TBS770 3x54W/827/865 HDF AC-MLO CVC) contained three fluorescent tubes of 54 W, of which two were 6500 K and one was 2700 K. All luminaires were equipped with an acrylate micro-lens optic cover to blend the two lamp types. Using a calibrated spectroradiometer (JETI Specbos 1201), the illuminance level, spectral power distribution (SPD) and colour rendering index (CRI) were measured at eye level (while seated, 1.20 m) aimed in the gaze direction of the participants. The CRI at 4000 K was R_a = 87. The SPD of the lighting in the 200 lux and the 1000 lux condition (4000 K) is depicted in Fig. 1.

During the practice phase and the baseline phase (first 15 min block), the ceiling mounted luminaires provided illumination of approximately 100 lux (4000 K) at eye level (photon density:

Table 1
Distribution of participants across n-back versions and timeslots.

	1-Back		2-Back		3-Back		
	Males	Females	Males	Females	Males	Females	Total
Morning timeslot 1 (9:00 AM)	0	3	2	4	2	1	12
Morning timeslot 2 (10:45 AM)	5	3	2	3	3	4	20
Afternoon timeslot 1 (12:15 PM)	2	1	3	1	2	2	11
Afternoon timeslot 2 (13:45 PM)	2	2	2	1	2	1	10
Afternoon timeslot 3 (15:45 PM)	2	1	1	2	2	3	11
Total	11	10	10	11	11	11	64

Table 2
Spectrally weighted α-opic illuminance levels for each lighting condition.

	λ_{max}	α-Opic lux value (200 lux)	α-Opic lux value (1000 lux)
Melanopsin	480.0	144	686
S-cone	419.0	147	686
M-cone	530.8	188	924
L-cone	558.4	204	1011
Rods	496.3	163	789

Note: α-Opic values for corneal spectral irradiance were determined using the calculation toolbox developed by Lucas et al. [44]. These values are based on healthy human eyes (32 years old, dilated pupils, 7 mm).

7.88×10^{13} photons s^{-1} cm^{-2}; irradiance: 30 μW/cm^2) and 275 lux horizontally at the table surface. During the experimental phase ceiling luminaires were set to 200 lux (4000 K) at eye level (photon density: 1.63×10^{14} photons s^{-1} cm^{-2}; irradiance: 61 μW/cm^2) and 580 lux at the table surface, or to 1000 lux (4000 K) at eye level (photon density: 8.09×10^{14} photons s^{-1} cm^{-2}; irradiance: 304 μW/cm^2) and 2900 lux at the table surface. Illuminance values at eye level for each of the retinal photoreceptors can be viewed in Table 2.

The room was set up in such a way that the illuminance levels at eye level in each of the workstations were as similar as possible. Divergences of the illuminance levels between the workstations were minimal: 8 lux at eye level in the baseline condition, 15 lux at eye level in the 200 lux condition and 87 lux at eye level in the 1000 lux condition. Participants were seated at the same workstation during both sessions.

The furnishing in each workstation consisted of a desk (1.4 m by 0.8 m) and a red chair. A 15.6-inch laptop was placed on each of the desks with a keyboard, mouse and headphones plugged in. Participants faced the wall opposite to the curtain. The task displayed at each of the laptops consisted of a grey background using a low but readable brightness level. The lighting measurements were performed with this background on the screen. The walls and ceiling of the laboratory room were off-white and had a reflectance of 87%, the floor was grey-blue with a reflectance of 19%, and the desk was light-grey with a reflectance of 39%. Windows in the room were fully covered using white screens; thus, there was no daylight contribution during the experiment.

2.4. Procedure

Before participating in the study, participants completed several online questionnaires probing person characteristics such as chronotype and light sensitivity. Participants were asked to keep their sleep-wake timing two days before each experimental session similar to their habitual sleep schedule on workdays as reported in the MCTQ (±30 min).

During the first visit to the laboratory, participants first signed an informed consent form and were then seated at their workstation. Each session started with instructions and a practice phase during which the performance tasks were practiced. After the practice phase, participants had the opportunity to ask any remaining questions about the study set-up. Subsequently, the baseline phase started where participants completed one measurement block of 15 min, which included the three performance tasks (*n*-back task, Forward Digit-Span Task and Backward Digit-Span Task) as well as subjective indicators of performance, mood, tension, vitality and sleepiness. After the baseline phase, illuminance levels were changed from 100 lux to either 200 lux or 1000 lux at eye level, after which participants engaged in four similar 15-min measurement blocks.

At the end of each session, participants completed a questionnaire on time of sleep onset and sleep offset, time spent outside, travelling time outside and caffeine and food consumption before the start of the experiment. In addition, participants evaluated the lighting in the room (e.g., whether it was relaxing or stimulating, dim or bright etc.). Finally, after the second session, participants completed a short questionnaire about their general beliefs regarding the effects of light on mood, vitality and performance. A full overview of the experimental procedure is depicted in Table 3. Participants received a 30-euro compensation after completing both sessions. The study was conducted from March to May 2014.

2.5. Measurements

2.5.1. Task performance

In each measurement block, two auditory working memory tasks (digit span tasks [45] and *n*-back task [46]) were employed to test the effects of illuminance level on cognitive performance. Auditory tasks were used because the study focused on NIF effects of light on cognitive performance. Visual tasks would be less appropriate because task performance might also depend on visual comfort and visual abilities. Each measurement block started with a 3.5-min *n*-back task. In this task, participants heard a sequence of one-syllable consonants (D, F, G, H, L, M, P, Q, S and W) and were asked to react as fast as possible to a consonant by pushing the spacebar in case the consonant was similar to the consonant *n* positions back. Three difficulty levels were used for the *n*-back task (1-, 2- and 3-back), which increasingly rely on more complex working memory updating abilities [46]. Single consonants were presented for roughly 800 ms through headphones using an inter-stimulus interval of 1000 ms. Mean reaction times (RTs) to target stimuli as well as accuracy scores (percentage of correct responses) were used as outcome measures for this task. *N*-back RTs were transformed to 1/RT outcome values to realize more normally distributed variables. Because of a ceiling effect on 1-back task accuracy scores (almost all participants had a score of 100%), only 2- and 3-back accuracy scores were analyzed.

The second and third task of each measurement block were the Forward and Backward Digit-Span Task (FDST and BDST). The FDST is described as a simple span task comprising storage and maintenance components of working memory [47]. The BDST is more complex in the sense that the stored information also has to be mentally reversed in working memory, which requires central executive functions [47]. In the current study, the FDST started at a length of four digits building up to a length of nine digits. It was

Table 3
Overview of one experimental session.

One full experimental session						
Practice-phase: 100 lux at eye level	Baseline-phase: 100 lux at eye level	Light exposure phase: 200 or 1000 lux at eye level				
Instructions and task practice	Task performance, subjective indicators	Block 1: similar to baseline	Block 2: similar to baseline	Block 3: similar to baseline	Block 4: similar to baseline	Questionnaires
10 min	15 min	15 min	15 min	15 min	15 min	5 min

decided to start at a length of four because a pilot study on this task conducted in our lab revealed that the three digit-span length was completed without any errors by all of the participants in the pilot. For the BDST, the starting length was three digits and the ending length was eight digits. For both tasks, participants completed two trials per digit-span length, resulting in 12 trials per task per measurement block. Total correct responses (i.e., correctly reported full digit-spans) per measurement block for digit-span lengths four to eight were used as the outcome variable. One participant was excluded for the FDST analyses because of zero scores (no response registration) during the first measurement block.

2.5.2. Subjective task performance

After each task, participants were asked to rate their performance on the task. This was assessed by four items: how well participants thought they had performed on the task, how motivated they were to perform the task as well as possible, how well they could concentrate on the task and how much mental effort they had to put into the task. These items were rated on a Visual Analogue Scale (VAS) ranging from 0 (not at all) to 100 (very much).

2.5.3. Subjective sleepiness (after each task)

Subjective sleepiness was examined with the Karolinska Sleepiness Scale (KSS; [48]). This measurement employs a 9-point scale using response options from 1 (extremely alert) to 9 (extremely sleepy–fighting sleep).

2.5.4. Subjective vitality, tension and mood (after each measurement block)

Subjective vitality and tension were assessed using six items adopted from the activation–deactivation checklist [49]. Four of these items belong to the vitality subscale (feelings of positive energy, alertness, sleepiness (reversed), and lacking energy (reversed); $\alpha = 0.84$) and two of these items belong to the tension scale (tension and calmness (reversed); $\alpha = 0.57$). Mood was assessed using single items for positive affect ('happy') and negative affect ('sad'). All items were rated on a Likert scale ranging from 1 (definitely not) to 5 (definitely).

2.5.5. Evaluation of the lighting

At the end of each session, participants were asked to evaluate the lighting in the room using six 5-point Likert-scale items which were adopted from Flynn et al. [50]. Three items focused on the experienced pleasantness of the lighting ('unpleasant – pleasant', 'uncomfortable – comfortable' and 'disturbing – not disturbing'; $\alpha = 0.85$). Furthermore, three items were used to assess experiences concerning the lighting colour ('warm–cold'), brightness ('dim–bright') and whether the lighting was activating ('relaxing–stimulating').

2.5.6. Possible confounding variables

Potential confounding variables were assessed before the start of the experiment. These consisted of chronotype, using the MCTQ [42]; Global sleep quality, using the Pittsburg Sleep Quality Index [51]; General fatigue, using the Checklist Individual Strength ($\alpha = 0.93$) [52]; Light sensitivity, using three items regarding light exposure sensitivity ($\alpha = 0.75$), and Introversion/Extraversion, using the extraversion subscale of the Dutch version of the Big Five Inventory ($\alpha = 0.81$) [53]. Furthermore, at the end of each session, session specific confounders were investigated (e.g., sleep quality, caffeine consumption, time spent outside etc.). Finally, at the end of the second session, participants were asked to rate the extent to which they thought light influences mood, performance, concentration, vitality, alertness and motivation on a 5-point Likert scale. These questions were only asked after the second session of the study in order to disguise the main aim of the study. However, during the debriefing after the second session, we noticed that participating in the study may have influenced participants' responses on these lighting beliefs questions. Therefore, we decided not to analyze this variable as a possible confounder.

3. Results

Because of the nested structure of the data, Linear Mixed Model (LMM) analyses were conducted to investigate the effects of illuminance level on repeated measurements of task performance and subjective indicators. For each of the outcome variables, the first LMM model that was tested included Lighting condition (200 lux vs. 1000 lux), Measurement block and Lighting condition × Measurement block as fixed factors. In case the effect of illuminance level was not moderated by time in session (Measurement block), the interaction term was excluded from the model and a second LMM was conducted including Lighting condition, Time of day and the interaction between Lighting condition and Time of day. Measurement block was included as a main effect in this model if it turned out to be significant. Changes in model fit were calculated based on maximum likelihood estimations.

Only statistics on main or interaction effects of Lighting condition were reported, statistics on main effects Measurement block were not reported. In case a significant interaction between Lighting condition and Measurement block was found, moderation of Time of day was investigated in the measurement block(s) revealing significant post hoc tests on the investigated outcome variable. To assess possible differential effects of illuminance level in the morning vs. the afternoon, the five timeslots were categorized as either morning (9:00 and 10:45 AM) or afternoon (12:15, 13:45 and 15:45 PM).

Pseudo and marginal R^2-values were calculated for models containing statistically significant main or interaction effects of Lighting condition. Pseudo R^2-values indicate the proportion of reduction in variance of residuals from the null-model to the final (full) model at level 2 [54]. Marginal R^2-values indicate the total variance that can be explained by all the fixed effects included in the final model [55].

3.1. Preparatory analyses

3.1.1. Baseline comparisons

For each of the outcome variables, potential baseline differences prior to the actual light manipulations were investigated. Near-significant baseline differences between lighting conditions emerged for 3-back task accuracy and for FDST and BDST accuracy. In subsequent analyses on these tasks, baseline scores were therefore included as a covariate.

3.1.2. Potential confounding variables

Preparatory LMM analyses were conducted to investigate within-subject differences in possible confounding variables between the two lighting conditions (hours of sleep, self-rated sleep quality on the night before the session, time awake, time spent outside and food and caffeine consumption during the hour before the session). No significant differences were found. These variables as well as individual characteristics (chronotype, general sleep quality, general fatigue and light sensitivity) were also checked for differences between morning and afternoon timeslots. Time spent outside and, as expected, Time awake significantly differed between morning and afternoon timeslots.

3.1.3. Manipulation check

A manipulation check was conducted for the *n*-back tasks and the digit-span tasks. It was found that, based on accuracy scores as well as RTs, the 1-back task was significantly easier than the 2-back

Table 4
Overview of light effects on *n*-back task performance in the morning and afternoon.

	Morning				Afternoon			
	200 lux	1000 lux	Statistics		200 lux	1000 lux	Statistics	
	EMM (*SE*)	*EMM* (*SE*)	*F* (*df*)	*p*	*EMM* (*SE*)	*EMM* (*SE*)	*F* (*df*)	*p*
2-Back accuracy	80.0% (3.8%)	85.1% (3.8%)	1.89 (1, 33)	0.18	**84.6% (4.0%)**	**75.8% (4.0%)**	**5.08 (1, 33)**	**0.03**
3-Back accuracy	54.6% (5.4%)	54.6% (5.2%)	0.00 (1, 31)	0.99	62.3% (4.8%)	60.3% (4.9%)	0.24 (1, 29)	0.63
1-Back 1/RTs	1.05 (0.04)	1.08 (0.04)	0.66 (1, 29)	0.42	1.07 (0.04)	1.05 (0.04)	0.11 (1, 29)	0.74
2-Back 1/RTs	0.95 (0.03)	0.97 (0.03)	0.54 (1, 28)	0.47	0.96 (0.03)	0.95 (0.03)	0.10 (1, 28)	0.76
3-Back 1/RTs	0.92 (0.03)	0.91 (0.03)	0.02 (1, 36)	0.88	0.91 (0.03)	0.92 (0.03)	0.42 (1, 36)	0.52

Note: *EMM* stands for estimated marginal mean and *SE* stands for standard error. Significant Lighting condition × Time of day interactions are indicated using bold fonts for the outcome measures in the first column. Significant post hoc differences are indicated in bold in the subsequent columns.

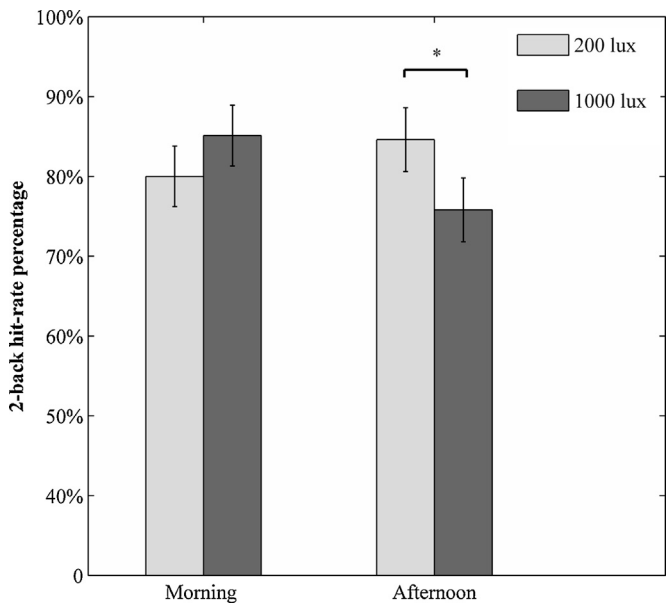

Fig. 2. 2-Back task hit-rate percentage for both lighting conditions in the morning and afternoon. Whiskers represent standard errors. $^{*}p < 0.05$.

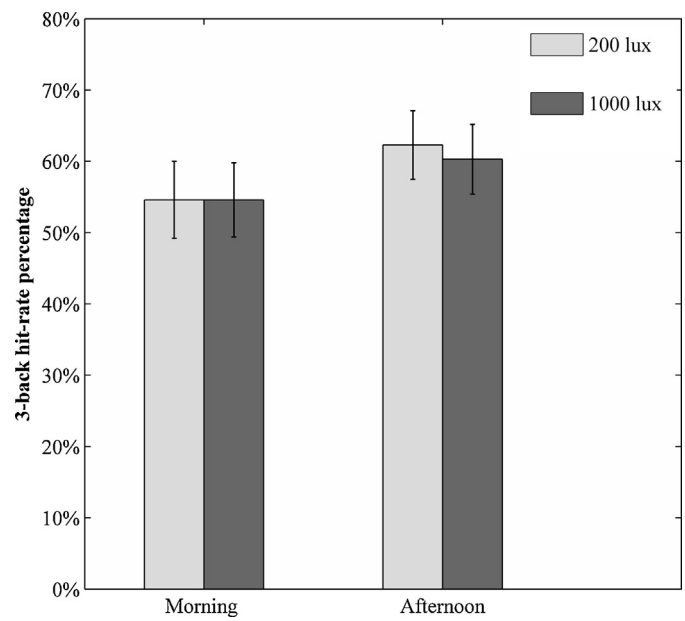

Fig. 3. 3-Back task hit-rate percentage for both lighting conditions in the morning and afternoon. Whiskers represent standard errors.

task and the 3-back task, and that the 3-back task was significantly more difficult than the 2-back task. In addition, the BDST was significantly more difficult than the FDST as shown by significantly lower accuracy scores for the BDST.

3.2. Effects on performance

3.2.1. 2- and 3-back task accuracy

LMM analysis for 2-back task accuracy did not yield a significant main effect for Lighting condition ($F(3,29) < 1$, *ns.*) nor a significant Lighting condition × Measurement block interaction ($F(3,108) < 1$, *ns.*). The LMM testing the Lighting condition × Time of day interaction significantly improved the model fit ($\chi^2(2) = 6.21$; $p = 0.04$). This model revealed a significant Lighting condition × Time of day interaction ($F(1,33) = 6.66$, $p = 0.02$). The model reduced the unexplained variance of 2-back accuracy scores with 10.0% (pseudo $R^2 = 0.100$) as compared to the null model. The three fixed effects in this model explained 3.3% of the total variance in 2-back accuracy scores (marginal $R^2 = 0.033$).

Post hoc inspection revealed that participants performed significantly worse on the 2-back task under 1000 lux than under 200 lux in the afternoon, but not in the morning (see Table 4). Fig. 2 depicts a bar chart of the percentage correct target responses for both lighting conditions in the morning and afternoon.

LMM analyses for the 3-back task accuracy revealed no significant main ($F(1,30) < 1$, *ns.*) or interaction effect of Lighting condition and Measurement block ($F(3,104) < 1$, *ns.*). The model investigating Time of day effects did not significantly improve the model fit ($\chi^2(2) = 1.14$, *ns.*), nor did it reveal any significant effects. Fig. 3 shows the 3-back task hit-rate percentages for both lighting conditions in the morning and afternoon.

3.2.2. N-*back task reaction times*

LMM analysis on 1-back task average inverse RTs did not yield a significant main or interaction effect of Lighting condition (p's > 0.37). The LMM including the Lighting condition × Time of day interaction did not significantly improve the model fit ($\chi^2(2) = 0.65$; *ns.*), nor did it reveal any significant effects (see Table 4).

A similar LMM was conducted for 2-back task RTs. Results revealed no significant main or interaction effect of Lighting condition (p's > 0.85). The LMM including the Lighting condition × Time of day interaction did not significantly improve the model fit ($\chi^2(2) = 0.64$; *ns.*), nor did it reveal any significant effects (see Table 4).

Finally, LMM analysis for the 3-back task RTs also showed no significant main or interaction effect of Lighting condition was (p's > 0.72). The LMM including the Lighting condition × Time of day interaction did not significantly improve the model fit ($\chi^2(2) = 0.30$; *ns.*), nor did it reveal any significant effects (see Table 4).

3.2.3. FDST performance

LMM analysis examining the total number of correctly reproduced forward digit-spans per measurement block revealed a

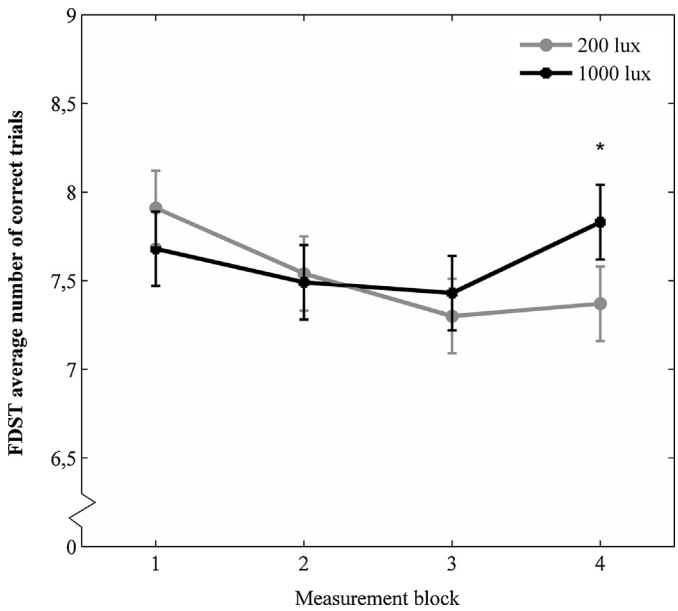

Fig. 4. FDST average number of correct trials for each of the four light exposure measurement blocks. Whiskers represent standard errors. $^{*}p < 0.05$.

Measurement block × Lighting condition interaction that did not reach statistical significance ($F(3,324) = 1.88$, $p = 0.13$). Since we expected to find more pronounced effects of illuminance level with increasing time in session (longer light exposure duration), it was decided to examine post hoc effects of Lighting condition for the final two measurement blocks using Bonferroni correction. These tests revealed that participants performed significantly better ($p = 0.03$) on the FDST in the 1000 lux condition ($EMM = 7.83$; $SE = 0.20$) than in the 200 lux condition ($EMM = 7.36$; $SE = 0.20$; see Fig. 4) in the final measurement block. No significant difference was found in the third measurement block (200 lux: $EMM = 7.29$; $SE = 0.20$; 1000 lux: $EMM = 7.44$; $SE = 0.20$; $p = 0.50$).

When focusing only on the final measurement block, an LMM including Lighting condition and FDST baseline scores as fixed factors rendered a significant main effect of Lighting condition ($F(1,60) = 4.87$, $p = 0.03$). This model reduced the unexplained variance of FDST accuracy in block four as compared to the null model with 4.3% (pseudo $R^2 = 0.043$). The fixed effects in this model explained 6.8% of the unexplained variance (marginal $R^2 = 0.068$).[1] Adding Time of day and Lighting condition × Time of day did not result in a significant improvement of the model ($\chi^2(2) = 0.89$; *ns.*). This test yielded a non-significant Lighting condition × Time of day interaction ($F(1,63) < 1$, *ns.*), while the effect of Lighting condition remained significant ($F(1,61) = 4.98$, $p = 0.03$). This indicates that the effect of illuminance level in measurement block four was independent of time of day. Table 5 shows the statistics for the post hoc comparisons of the Lighting condition × Time of day interaction effect on FDST performance for the final measurement block. These results indicate that participants benefitted from bright light in the morning as well as the afternoon, but that this effect was most pronounced in the afternoon.

3.2.4. BDST performance

LMM analysis examining the total number of correctly reproduced backward digit-spans per measurement block yielded no significant main effect of Lighting condition, nor a significant interaction between Lighting condition and Measurement block (both p's > 0.69).

The LMM model investigating the interaction between Lighting condition and Time of day showed no significant improvement in the model fit ($\chi^2(2) = 4.56$; $p = 0.10$). However, LMM analyses revealed a significant Lighting condition × Time of day interaction ($F(1,97) = 4.38$, $p = 0.04$). This model reduced the unexplained variance of BDST accuracy of the null model with 3% (pseudo $R^2 = 0.03$). The fixed effects in this model explained 17.3% of the unexplained variance (marginal $R^2 = 0.173$).[2] The pattern of responses of this interaction suggested somewhat better performance in the 1000 lux condition than in the 200 lux condition in the morning, and the reverse – i.e., worse performance in the 1000 lux condition as compared to the 200 lux condition – in the afternoon, although the post hoc comparisons did not reach significance (see Table 5). Fig. 5 shows bar charts for the effect of illuminance level on BDST performance in the morning and afternoon.

3.2.5. Summary of lighting effects on performance indicators

Participants performed significantly better on the FDST under 1000 as compared to 200 lux during the final measurement block. For accuracy on the BDST and the 2-back task, a different pattern was found: although participants performed slightly better (not significant) under 1000 lux in the morning, they performed worse on these tasks in the afternoon under 1000 as compared to 200 lux. These effects were independent of measurement block. With respect to accuracy on the 3-back task, as well as reaction time performance on the 1-, 2- and 3-back task, no effects of illuminance level were found.

3.3. Effects on subjective indicators

3.3.1. Sleepiness

Sleepiness ratings were reported after each task in every measurement block, resulting in 12 repeated measures. LMM analysis using a full factorial model including Lighting condition, Measurement Block and Task type (n-back vs. FDST vs. BDST) did not reveal a main effect of Lighting condition ($F(1,64) < 1$, *ns.*) nor an interaction between Lighting condition and Task type ($F(2,1408) = 1.22$, *ns.*) and Lighting condition and Measurement block ($F(3,1408) < 1$, *ns.*).

LMM analysis investigating time-of-day differences revealed an interaction effect between Lighting condition and Time of day that approached significance ($F(1,381) = 3.42$, $p = 0.07$). Post hoc results revealed that participants reported significantly less sleepiness in the 1000 lux condition as compared to the 200 lux condition in the morning (see Table 6). In the afternoon, no significant difference in sleepiness between the lighting conditions was found (see Table 6).

3.3.2. Vitality, tension and mood

For vitality, tension and mood measures, completed at the end of each measurement block, an LMM analysis was conducted using Lighting condition and Measurement block and the interaction between the two as fixed factors. These models did not reveal significant main effects of Lighting condition nor significant interactions between Lighting condition and Measurement Block (all p's > 0.35). LMM analysis testing time-of-day effects revealed a near-significant interaction effect between Lighting condition and Time of day for subjective feelings of vitality ($F(1,77) = 3.93$, $p = 0.05$). The results suggested more vitality under bright light in the morning and less vitality under bright light in the afternoon,

[1] Note that this model also contains FDST accuracy baseline scores. The model only containing the baseline scores as a fixed factor explained 5.3% of the variance.

[2] Note that this model also contains BDST accuracy baseline scores. The model only containing the baseline scores as a fixed factor explained 14.1% of the variance.

Table 5
Overview of light effects on digit-span task performance in the morning and afternoon.

	Morning				Afternoon			
	200 lux	1000 lux	Statistics		200 lux	1000 lux	Statistics	
	EMM (*SE*)	*EMM* (*SE*)	*F* (*df*)	*p*	*EMM* (*SE*)	*EMM* (*SE*)	*F* (*df*)	*p*
FDST accuracy	7.40 (0.28)	7.68 (0.28)	0.882 (1, 61)	0.35	**7.31 (0.29)**	**7.99 (0.29)**	**4.67 (1, 62)**	**0.03**
BDST accuracy	6.03 (0.20)	6.26 (0.20)	1.23 (1, 94)	0.27	6.22 (0.21)	5.84 (0.21)	3.38 (1, 100)	0.07

Note: For FDST performance, only the final measurement block was taken into account. *EMM* stands for estimated marginal mean and *SE* stands for standard error. Significant Lighting condition × Time of day interactions are indicated using bold fonts for the outcome measures in the first column. Significant post hoc differences are indicated in bold in the subsequent columns.

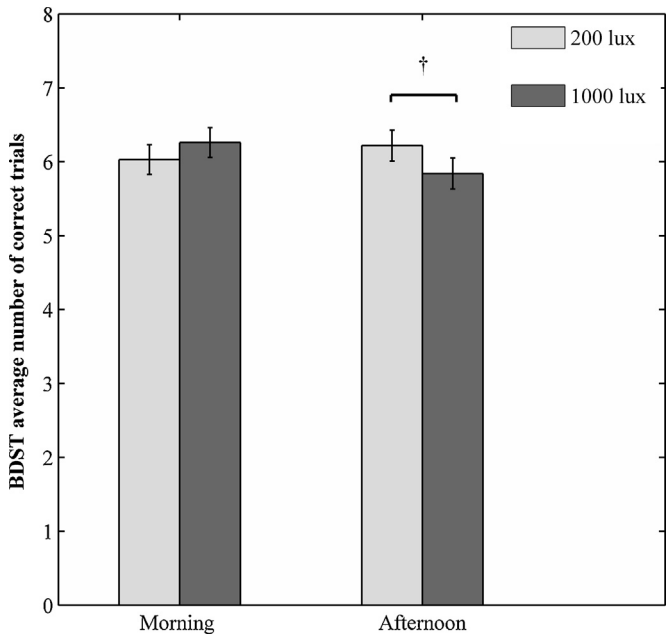

Fig. 5. Lighting condition × Time of day interaction for BDST average accuracy over the four measurement blocks. Whiskers represent standard errors. †$p < 0.1$.

but post hoc comparisons revealed no significant differences (both p's > 0.16; see Table 6).

No significant interaction between Lighting condition and Time of day was found for feelings of tension ($F(1,90) = 2.61$, $p = 0.11$) and mood (positive affect: $F(1,81) = 2.16$, $p = 0.15$; negative affect: $F(1,83) < 1$, *ns.*), see Table 6.

3.3.3. Subjective performance

LMM analyses using a full factorial model with Lighting condition, Measurement block and Task type were conducted to investigate the effect of illuminance level on each of the four subjective performance ratings. The results revealed a significant Lighting condition × Measurement block interaction ($F(3,1408) = 2.79$, $p = 0.04$) for self-reported ability to concentrate, which was independent of Task type (non-significant Lighting condition × Measurement block × Task type interaction: $F(14,1408) < 1$, *ns.*). The interaction suggested somewhat better subjective concentration during the final two measurement blocks under 1000 lux (block 3: *EMM* = 63.23, *SD* = 2.13; block 4: *EMM* = 63.52, *SD* = 2.13) as compared to 200 lux (block 3: *EMM* = 61.66, *SD* = 2.13; block 4: *EMM* = 60.76, *SD* = 2.13), while subjective concentration was better under 200 lux (block 1: *EMM* = 70.54, *SD* = 2.13; block 2: *EMM* = 67.19, *SD* = 2.13) as compared to 1000 lux (block 1: *EMM* = 68.71, *SD* = 2.13; block 2: *EMM* = 65.32, *SD* = 2.13) in the first two measurement blocks. However, post hoc comparisons using Bonferroni correction revealed no significant differences between lighting conditions in any of the measurement blocks (all p's > 0.15).

LMM analysis examining potential time-of-day dependent effects did not yield an improvement in the model fit ($\chi^2(2) = 0.92$; *ns.*) and revealed a non-significant Lighting condition × Measurement block × Time of day interaction ($F(7,592) = 1.07$, *ns.*). In this model, the Lighting condition × Measurement block interaction remained significant ($F(3,1408) = 2.78$, $p = 0.04$).

No significant main or interaction effects of Lighting condition were found on how well participants thought they had performed, how motivated they were to perform as well as possible, and how much mental effort was needed (all p's > 0.19).

3.3.4. Subjective appraisals of the lighting

LMMs using Lighting condition as a fixed factor and experienced pleasantness, brightness, activation level and colour of the lighting as outcome variables revealed a significant main effect of Lighting condition for pleasantness ($F(1,64) = 11.12$, $p < 0.01$), brightness ($F(1,64) = 34.72$, $p < 0.01$) and level of activation ($F(1,64) = 13.60$, $p < 0.01$), but not for the experienced colour of the lighting ($F(1,64) = 1.07$, $p = 0.31$). Post hoc comparisons showed that participants evaluated the 1000 lux condition as less pleasant (*EMM* = 9.25; *SE* = 0.33), brighter (*EMM* = 4.41; *SE* = 0.09) and more activating (*EMM* = 3.77; *SE* = 0.10) than the 200 lux condition (pleasantness: *EMM* = 10.48; *SE* = 0.33; brightness: *EMM* = 3.68; *SE* = 0.09; activation level: *EMM* = 3.34; *SE* = 0.10). When testing potential time-dependent effects for the lighting appraisals, no

Table 6
Overview of light effects on subjective sleepiness vitality, tension and mood in the morning and afternoon.

	Morning				Afternoon			
	200 lux	1000 lux	Statistics		200 lux	1000 lux	Statistics	
	EMM (*SE*)	*EMM* (*SE*)	*F* (*df*)	*P*	*EMM* (*SE*)	*EMM* (*SE*)	*F* (*df*)	*p*
Sleepiness	**5.57 (0.22)**	**5.30 (0.22)**	**4.29 (1, 381)**	**0.04**	4.75 (0.22)	4.83 (0.22)	0.29 (1, 380)	0.59
Vitality	2.98 (0.11)	3.15 (0.11)	2.05 (1, 77)	0.16	3.33 (0.11)	3.18 (0.11)	1.88 (1, 77)	0.17
Tension	2.23 (0.11)	2.19 (0.11)	0.21 (1, 90)	0.65	1.95 (0.11)	2.10 (0.11)	3.35 (1, 90)	0.07
Happy	2.71 (0.15)	2.68 (0.15)	0.12 (1, 81)	0.73	2.60 (0.15)	2.76 (0.15)	3.00 (1, 81)	0.09
Sad	1.28 (0.08)	1.30 (0.08)	0.08 (1, 83)	0.78	1.28 (0.08)	1.26 (0.08)	0.08 (1, 83)	0.78

Note: *EMM* stands for estimated marginal mean and *SE* stands for standard error. Significant Lighting condition × Time of day interactions are indicated by bold labels in the first column. Significant post hoc differences are indicated in bold in the subsequent columns.

significant Lighting condition × Time of Day interactions were found (all p's > 0.16), indicating that these effects were independent of time of day.

3.3.5. Summary of lighting effects on subjective indicators

Participants felt significantly less sleepy under 1000 lux as compared to 200 lux in the morning but not in the afternoon. In addition, participants felt somewhat more vital under 1000 lux as compared to 200 lux in the morning, but slightly less vital under 1000 lux as compared to 200 lux in the afternoon. No significant effect of illuminance level was found on feelings of tension and mood. Participants evaluated the 1000 lux condition as significantly less pleasant, brighter and more activating as compared to the 200 lux condition (regardless of time of day). Finally, a higher illuminance level seemed to lead to somewhat better subjective concentration levels during the final two measurement blocks, but did not affect other subjective task performance measures.

4. Discussion

The current study examined the NIF effects of illuminance level on working memory performance using two tasks varying in difficulty level. To our knowledge, this was the first study to systematically examine whether differences in the effects of light intensity on cognitive performance may be explained by task difficulty. The second aim of this study was to examine effects of illuminance level on subjective performance, sleepiness, vitality, tension and mood. Based on previous research we hypothesized that a higher illuminance level would be more beneficial for performance on relatively easy tasks as compared to more difficult tasks. In addition, we expected subjective vitalizing effects under a higher illuminance level.

In order to test these hypotheses, participants were exposed to illuminance levels of 200 and 1000 lux (on separate occasions) in a within-subject design while performing two versions of the digit-span task (difficulty level manipulated within subjects) and one of the three versions of the n-back task (difficulty level manipulated between subjects). In addition to performance measures, subjective measures of performance, sleepiness, vitality and tension and mood were repeatedly examined during the 60-min light exposure. For each of the outcome variables, it was investigated whether the effect of illuminance level was moderated by time in session and/or time of day.

4.1. Performance measures

Overall, we found evidence that performance on the (easier) FDST benefitted more from bright light exposure than performance on the (more difficult) BDST. For the FDST, the improvement in performance under bright light was found only during the final measurement block. Participants benefitted from bright light during FDST performance in the morning as well as the afternoon, but this effect was most pronounced during the afternoon. In contrast, we found slightly worse performance on the BDST during the afternoon, which was independent of time in session. In line with our hypothesis, illuminance level thus differentially affected performance on these two tasks in the afternoon, leading to better performance on the easier task and somewhat worse performance on the more difficult task.

With respect to the n-back task however, we did not find evidence for more beneficial effects of bright light exposure on easier as compared to more complex task versions. More specifically, the only effect of illuminance level on n-back task performance was found for accuracy on the 2-back task. On this version of the n-back task, a similar but more pronounced pattern was found as for the BDST; participants performed significantly worse under bright light exposure during the afternoon. No significant effect of illuminance level was found for 3-back task accuracy nor on reaction times on any of the n-back versions. In case physiological arousal levels may partly underlie differences in performance under different lighting conditions (based on the YDL), one would expect participants to also perform worse on the relatively difficult 3-back task in the 1000 lux condition (at least in the afternoon). Thus, findings on the n-back task did not confirm our hypothesis regarding difficulty level. Unfortunately, because of a ceiling effect, it was not possible to test the effects of illuminance level on 1-back task accuracy and compare these to the effects found on the 2-back accuracy. In addition, it should be noted that the n-back task was manipulated between participants, leading to less power to detect relatively subtle effects of illuminance level on cognitive task performance as compared to the digit-span task.

The results on performance measures are partly in line with those found in a recent study by Smolders et al. [16]. Their results revealed an overall effect of illuminance level on accuracy on a visual 2-back task, indicating that participants performed better under 200 lux as compared to 1000 lux at eye level [16]. However, time of day was not taken into account in the analysis of that study, so it is unclear whether participants were differentially affected by illuminance level in the morning vs. the afternoon. It is possible that the overall effect found in the study of Smolders et al. [16] resulted from worse performance only in the afternoon timeslots, as was found in the current study. Findings on 2-back reaction times in the study Smolders et al. also agreed with the findings in the current study, showing no effect of illuminance levels on reaction times of response targets [16]. In contrast with our results, the study of Santhi et al. revealed that blue-enhanced lighting conditions (750 lux at 7280 K and 195 lux 7717 K), as compared to non-blue-enhanced lighting conditions (19 lux at 2592 K and 200 lux at 2529 K), significantly improved response times on the 3-back task towards the end of 4 h light exposure. However, as in our study, accuracy scores on the 3-back task were unaffected by the lighting conditions [22]. Apart from differences in colour temperature of the light sources, differences in exposure duration may explain the divergent results on 3-back reaction times.

Last, a recent study investigated the effect of morning dawn simulation (MDS; increasing illuminance level from 0 to 250 lux at eye level 30 min before wake-up time) on, amongst others, 1-, 2- and 3-back task accuracy in sleep restricted subjects [21]. As compared to a control condition receiving low lighting levels at wake-up (<8 lux at eye level), this study revealed improvements in 1-back accuracy performance about 4 h after MDS and on 3-back task performance directly after MDS. No effect of the lighting manipulation was found on 2-back task performance. It is, however, not possible to directly compare the findings of Gabel et al. to those of the current study as subjects were sleep restricted and lighting conditions differ in timing, intensity, temporal pattern and duration. Nevertheless, these results also suggest a moderating role of task difficulty in NIF effects of light on working memory performance. Yet, similar to the current n-back accuracy results, their findings are not consistent with the hypothesized role of task difficulty in the relationship between arousal and performance conform the YDL.

4.2. Subjective indicators

With respect to the subjective indicators, participants reported to feel significantly less sleepy under bright light during the morning timeslots, while illuminance did not affect sleepiness ratings in the afternoon. For vitality, we found a near-significant interaction effect between illuminance level and time of day, suggesting that bright light may – in line with the results on sleepiness – be beneficial for feelings of vitality in the morning but not in the afternoon.

Previous studies revealed consistent improvements on KSS ratings under bright vs. dim light exposure in the afternoon [11,12]. However, in these studies participants were light- and/or sleep-deprived before the start of the light exposure manipulation, and the dim light control conditions were much lower than in the current study (<10 lux). Therefore, results of these studies may have resulted in larger positive effects of illuminance level on sleepiness measures in the afternoon. Furthermore, in contrast to the results reported here, the study of Smolders et al. found that the effects of illuminance level on sleepiness and vitality were independent of time of day [13]. Since the design and lighting manipulation of the study of Smolders et al. were very similar to the current study it is difficult to determine what factors may have caused these differential effects. Seasonal differences may play a role in explaining the divergence in results. The current study took place in early spring while the study of Smolders et al. was conducted during the summer period. In addition, the tasks used in the current study had a relatively high cognitive load as compared to the tasks used in the study of Smolders et al. Possibly, these two aspects led to different effects of illuminance level on sleepiness and vitality.

Similar as in a previous study with a comparable lighting manipulation [13], we found no significant effect of illuminance level on subjective tension, mood and subjective performance. Effects of bright light on mood are likely manifested over a longer exposure period or when exposure is repeated during multiple days as in bright light therapy [56]. A similar hypothesis could hold for subjective performance indicators. That is, if illuminance level does affect performance abilities, participants may only notice this after a longer exposure period or repeated daily exposure.

4.3. Time of day effects

One quite consistent finding in the current study is the role of time of day (morning vs. afternoon) as moderating variable in the effects of illuminance level on subjective and performance indicators. On the subjective indicators, bright light reduced subjective sleepiness in the morning but not in the afternoon. In addition, subjective feelings of vitality were somewhat higher under bright vs. dim light in the morning, while the opposite was true in the afternoon. For the performance indicators, we see performance deterioration on two relatively difficult tasks in the afternoon under 1000 as compared to 200 lux, while there is no effect of illuminance level in the morning on these tasks. On the other hand, we see an improvement on an easier task under 1000 as compared to 200 lux in the afternoon. Overall, we cannot draw a consistent conclusion about the role of time of day in these effects. Yet, it seems to be the case that bright light does not always lead to positive NIF effects when time of the day is considered. Of course, it should be noted that in the current study timeslots in the morning and afternoon were combined, which led to an aggregation of participants varying in internal time and different levels of homeostatic sleep pressure. Future studies should investigate the effects of illuminance level for separate timeslots (i.e., in the early and late morning, after lunch, and late in the afternoon) in participants with a similar internal time. This would, however, require more participants per timeslot than in the current study in order to have sufficient power to detect differences in performance between the lighting conditions.

The divergent results on performance and subjective measures in the morning and afternoon in the current study might be explained by the fact that that the light exposure history was quite different for the morning vs. afternoon sessions. In fact, participants reported that they had spent significantly more time outside when they came to the lab during the afternoon hours, which suggests that participants had been exposed to more light prior to the afternoon than the morning sessions. Prior light history is known to be an important predictor of the acute NIF effects of light on alertness and performance [57,58]. That is, if the prior light history is less bright and less blue-enriched, participants tend to respond more strongly in terms of alertness and performance to current bright and/or blue-enriched lighting conditions. In this line of reasoning, however, it would be expected that FDST performance in the current study would be better under bright light during the morning timeslots, while this was in fact more pronounced during the afternoon timeslots. In addition, current findings would suggest that this change in light sensitivity over the day could lead to both beneficial as well as detrimental effects of bright light exposure on performance and subjective measures. In future studies, it would be of interest to examine whether prior light history can indeed explain variations of NIF effects of illuminance throughout the day.

4.4. Underlying mechanisms

In the introduction, we discussed possible underlying mechanisms that may explain the effects of illuminance level on cognitive performance during the biological daytime. One suggestion was that light-induced modulation of task-specific brain networks could temporarily facilitate improved cognitive performance. This model, however, would be hard to reconcile with the current study's findings that performance sometimes deteriorated under bright light exposure. On the other hand, it is also possible that activation of the brainstem affects physiological arousal levels, which in turn affect performance abilities (as explained by the YDL). Indeed, previous studies confirmed that bright light, as compared to more dim light, increased arousal levels [10,28,29]. This line of reasoning could explain performance deterioration on complex tasks under brighter light because of increases in physiological arousal levels. In our study, bright light exposure may have influenced participants' state arousal levels during the afternoon in such a way that it negatively affected 2-back and, to a lesser extent, BDST performance, while arousal levels seemed to be more optimal for FDST performance. This was, however, not confirmed by 3-back task performance which was unaffected by illuminance level. A limitation of the current study is that no physiological measures were used to examine actual physiological arousal levels. For future studies, it would be of great value to include objective measures of arousal levels in order to examine whether differences in arousal level may explain differential effects of illuminance level on cognitive performance.

Overall, it can be concluded that the effects of illuminance level on performance and subjective indicators found in this study are quite subtle with small effect sizes. In addition, some effects were close to, but not statistically significant (such as the difference in BDST performance between lighting conditions in the afternoon). One potential reason for this is that the relative difference between illuminance levels reaching the retina between the two conditions is quite small. As can be seen in Table 2, the α-opic lux levels (dilated pupils) for melanopsin are about 145 lux for the 200 lux condition and 685 lux for the 1000 lux condition. However, since the pupil was more constricted in the 1000 lux as compared to the 200 lux condition as we did not use dilating eye drops, the total number of photons reaching the retina in the bright light condition is relatively smaller. Even more complicated is the fact that pupil constriction is also determined by individual physiological arousal levels, cognitive effort and interest [59,60], which likely leads to some individual variation in retinal exposure within lighting conditions. However, since the purpose of our research was translational, we chose illuminance levels that are realistic for an office environment, taking into account energy expenditure, human needs (including visual and nonvisual effects of light), and natural human responses. Most previous studies investigating the NIF effects of light on alertness and performance used more extreme lighting conditions, which may explain larger and more statistically significant effects found

in these studies (e.g., [10–12]). Nevertheless, the finding that a relatively small difference in illuminance level already influences cognitive performance and feelings of alertness can be of great importance for designing intelligent healthy and vitalizing light applications in for example offices and school buildings.

Our hypothesis that easier tasks would benefit more from bright light than more complex tasks was only partly supported. Obviously, more studies are needed to investigate this hypothesis, and test whether it also applies to tasks involving other cognitive abilities. It would also be of great interest to develop a dose–response curve for NIF effects of indoor lighting for subjective feelings of alertness and cognitive performance in day-active persons. To realize this, different realistic indoor illuminance levels and colour temperatures should be investigated, taking into account at least the timing of the exposure, so that optimal lighting characteristics for NIF effects can be determined. Furthermore, in order to investigate possible mediating mechanisms of NIF effects of light on cognitive performance, it is recommended to use both subjective as well as objective (physiological) measures of arousal levels.

5. Conclusion

In sum, the results of the current study partly confirm our hypothesis that task difficulty may play a role in the effect of illuminance level on cognitive task performance. Performance on the FDST improved under bright light exposure, while performance decreased slightly on the BDST. This difference in effect of illuminance level on digit-span performance only appeared in the afternoon, not in the morning. For the *n*-back task, our hypothesis was not confirmed. Bright light only negatively affected 2-back performance in the afternoon, while no effect of illuminance level was found for 3-back performance. Future studies should investigate whether task difficulty level may explain differential effects of illuminance level on performance on other types of cognitive tasks.

Time of day also seemed to be an important predictor in explaining variations in NIF effects of illuminance level. Several subjective as well as performance indicators were differentially affected by illuminance level in the morning vs. the afternoon. Although the current study did not reveal a consistent pattern of time of day differences in NIF effects, it can be concluded that, in the afternoon hours, bright light may not always be beneficial for NIF effects on performance. While previous studies largely focused on positive NIF effects of bright vs. dim and/or 'regular' light on performance, the current results suggest that, under certain circumstances, these effects can be detrimental. Since we know only very little about the NIF effects of light on subjective and objective indicators of alertness during the biological day, future studies should try to establish more insight in the variables, such as time of day, prior light history, physiological arousal and task characteristics, that may influence the direction and magnitude of these effects.

Conflicts of interest

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